

A Domain-Specific Computer Language for Market Monitoring: the Missing Link Between Thoughts and SAS®

Eric C. Grasby, SPP; Daniel Berleant, UA Little Rock

ABSTRACT

This paper presents a proof-of-concept design and implementation of a domain-specific language (DSL) for use in energy market monitoring. Energy markets are complex systems that enable market participants to economically trade energy at wholesale rates. Market monitors analyze these markets and screen for manipulative behavior, often using SAS for data extraction, transformation, and analysis.

While SAS is a well-established platform for working with data, many market monitors are not expert computer programmers. DSLs enable non-programmers to solve problems without requiring a background in software development. This paper presents a DSL that aims to alleviate this limitation by enabling a monitoring analyst to specify data analyses in an intuitive way that better suits their background and training. It then transpiles their output into valid SAS code, leveraging SAS data sets for data management and SAS procedures, DATA step programming, and SAS macros to handle complex functionality.

This paper, additionally, demonstrates the flexibility of this proposed design by providing example scripts that satisfy several use cases of this system. These use cases were identified by an anonymous panel of market monitors as part of a Delphi study.

INTRODUCTION

Motivation and context

Statement of contributions

BACKGROUND

Energy market structure and monitoring

The business problem: why a domain-specific language?

Why SAS?

REQUIREMENTS SOLICITATION

Delphi study methodology

To identify relevant use cases during the implementation of this DSL, the researchers chose to proctor a Delphi study. Eligibility to participate in this study was determined by only two (2) requirements:

1. The participant must be over the age of 18

2. The participant must be a member of the Energy Intermarket Surveillance Group (EISG)¹

These eligibility requirements were introduced to both adhere to responsible research ethics² and to establish a definition of “expert opinion” provided by each participant. Plainly, this ensured that the survey questions posed in this study were answered by the same individuals that may use a purpose-made DSL for energy market analytics. The study was conducted in two (2) rounds, where a round corresponds to a group of survey questions that were provided to the participants. These included a list of free-form text responses, multiple choice questions, and questions provided in a Likert format.

Panel Composition

At the time of administering this study, the EISG held a roster of 131 active members. Round #1 yielded a roughly 15% response rate (19 confirmed individuals participated in the first round). Only 11 of the original 19 participants elected to respond to round #2 (roughly a 58% retention rate of the original participants). Of the panelists, 16 of the EISG members were US-based market monitors. The remaining 3 worked for an internationally based market monitoring unit. Figure 1 professional role mix of the panelists, indicating that market monitoring analysts make up the largest portion of the panelists, with a mix of other roles including management, economists, data scientists, and a developer.

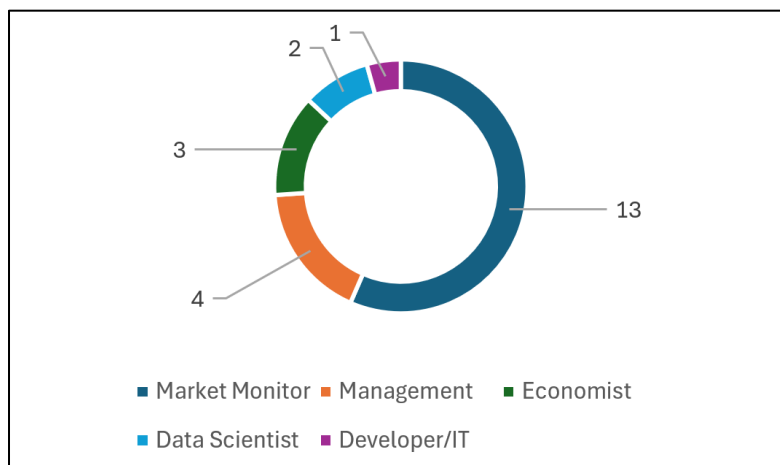


Figure 1 - Professional Role Mix of Delphi Panel

¹ The Energy Intermarket Surveillance Group (EISG) is the international, professional, conference of energy market monitors. Any given professional is a member of the EISG by virtue of working for an MMU that is an organizational member of the EISG.

² The research protocol for this Delphi study was approved by the UA Little Rock Institutional Review Board. The research protocol and copy of the waiver of informed consent can be found in [this](https://github.com/echarlesgrasby/dissertation/tree/master/ext) GitHub repository: <https://github.com/echarlesgrasby/dissertation/tree/master/ext>.

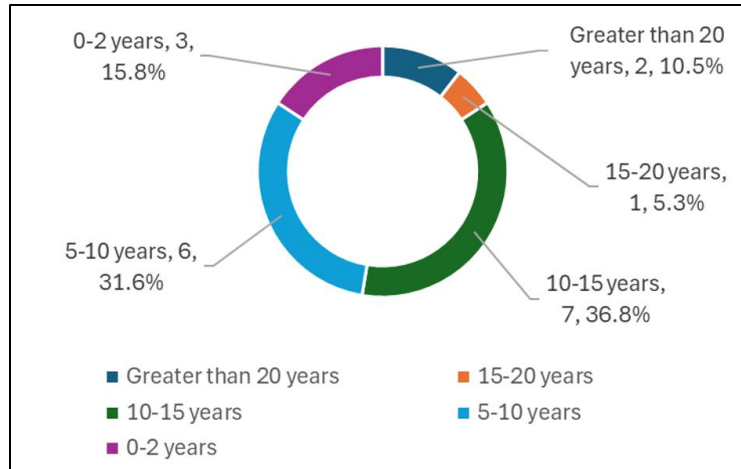


Figure 2 - Tenure Distribution of Delphi Panel

Results and thematic analysis

Derived requirements and use cases

DSL DESIGN

Design goals and constraints

Grammar specification

Example DSL statements

IMPLEMENTATION

To implement this domain-specific language, the researchers used two (2) main technologies: Python (including the Lark parser framework) and the SAS® OnDemand for Academics (SODA) environment.

Parser and transpiler architecture

Transpilation strategy

Handling complex functionality

EVALUATION AND WALKTHROUGH

Use case demonstrations

Verification of correctness

Coverage of derived requirements

CONCLUSION

Summary of Contributions

Limitations

Given that this paper provides a proof-of-concept for this DSL, there are several limitations with the implementation of this solution, namely:

1. The execution model of this DSL exists solely as preprocessor directives
2. Use of the resultant SAS code requires manual intervention

Future Work

REFERENCES

References are in-progress

ACKNOWLEDGMENTS

We would like to acknowledge both SPP, the Energy Information Authority (EIA), and DegreeDays.net for providing sources of publicly available data which made the construction and evaluation of this project possible. Readers are encouraged to visit these organization's respective websites for more information on public energy data:

- SPP (née Southwest Power Pool) -- <https://portal.spp.org/groups/integrated-marketplace>
- EIA -- <https://www.eia.gov/electricity/>
- BizEE Degree Days -- <https://www.degreedays.net/>

RECOMMENDED READING

Recommended reading in progress... Steven Stoft's *Power System Economics* at a bare minimum.

CONTACT INFORMATION

Your comments and questions are valued and encouraged. Contact the author(s) at:

Eric C. Grasby

Senior Market Monitor, Southwest Power Pool, Inc.
Doctoral Candidate, UA Little Rock, Department of Information Science
ecgrasby@ualr.edu
<https://linkedin.com/in/echarlesgrasby>

Dr. J. Daniel Berleant

Professor, UA Little Rock, Department of Information Science
jdberleant@ualr.edu
<https://www.linkedin.com/in/daniel-berleant-4938451/>

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